SMART PARKING MANAGEMENT SYSTEM (SPMS)

Requirement Report

Derived from the provided SPMS UML Use Case Diagram

Project: Smart Parking Management System

Document: Requirement Report

Basis: Provided UML Use Case Diagram

This report captures the actors, use cases, relationships, and functional/non-functional requirements represented or directly implied by the supplied use case diagram.

1. Project Overview

The Smart Parking Management System (SPMS) is a software system intended to support parking discovery, slot reservation, check-in/check-out, parking fee calculation, payment, occupancy monitoring, and administrative management of parking facilities.

The supplied UML diagram shows four actors: Driver, Admin, Sensor/Camera (IoT/ANPR), and Payment Gateway. Therefore, this requirement report retains those integrations as shown in the diagram rather than assuming a purely software-only implementation.

2. Actors

Actor	Role / Responsibility
Driver	Primary system user who searches for parking, views slot information, reserves parking, checks
Admin	Manages parking areas, parking slots, pricing, reservations, occupancy information, statistics, a
Sensor / Camera (IoT / ANPR)	External detection component that detects vehicle entry/exit events and supports updating park
Payment Gateway	External payment service involved in processing parking payments.

3. Use Cases

ID	Use Case	Primary Actor	Description
UC-01	Register / Login	Driver	Account registration and authentication.
UC-02	Manage Profile	Driver	View/update driver profile information.
UC-03	Search Parking	Driver	Search for suitable parking facilities.
UC-04	View Slot Details	Driver	View information about a selected parking slot.
UC-05	Check Parking Availability	Driver	Check whether parking capacity/slots are available.
UC-06	Reserve Parking Slot	Driver	Reserve an available parking slot.
UC-07	Cancel Reservation	Driver	Cancel an existing eligible reservation.
UC-08	Check-in	Driver	Record the beginning of a parking session.
UC-09	Check-out	Driver	Record the end of a parking session.
UC-10	Make Payment	Driver	Initiate payment for parking charges.
UC-11	View Booking History	Driver	View previous/current booking information.
UC-12	Receive Notifications	Driver	Receive system notifications related to parking activity.
UC-13	Generate Digital Receipt	System	Generate a digital receipt associated with payment.
UC-14	Process Payment	Payment Gateway	Process the parking payment transaction.
UC-15	Calculate Parking Fee	System	Calculate the applicable parking fee.
UC-16	Monitor Occupancy	Admin	Monitor current parking occupancy.
UC-17	View Revenue Reports	Admin	View reports related to parking revenue.
UC-18	View Parking Statistics	Admin	View parking usage/statistical information.
UC-19	Manage Parking Areas	Admin	Create/update/manage parking areas.
UC-20	Manage Parking Slots	Admin	Create/update/manage parking slots.
UC-21	Manage Pricing	Admin	Configure parking pricing.
UC-22	Manage Reservations	Admin	View and manage reservations.
UC-23	Detect Vehicle Entry / Exit	Sensor / Camera (IoT / ANPR)	Detect vehicle entry and exit events.
UC-24	Update Parking Occupancy	System	Update occupancy information based on vehicle movement.

4. Use Case Relationships

Base Use Case	Relationship	Included Use Case
Make Payment	<<include>>	Generate Digital Receipt
Make Payment	<<include>>	Process Payment
Reserve Parking Slot	<<include>>	Check Parking Availability
Check-out	<<include>>	Calculate Parking Fee
Detect Vehicle Entry / Exit	<<include>>	Update Parking Occupancy

The <> relationships indicate functionality that is required as part of the corresponding base use case.

5. Functional Requirements

5.1 Authentication and Profile

• FR-01: The system shall allow a Driver to register and log in.

• FR-02: The system shall authenticate valid user credentials before granting access to protected functions.

• FR-03: The system shall allow a Driver to manage profile information.

5.2 Parking Search and Availability

• FR-04: The system shall allow a Driver to search for parking.

• FR-05: The system shall allow a Driver to view parking slot details.

• FR-06: The system shall provide parking availability information.

• FR-07: The system shall check parking availability before a reservation is confirmed.

• FR-08: The system shall prevent reservation of an unavailable slot.

5.3 Reservation Management

• FR-09: The system shall allow a Driver to reserve an available parking slot.

• FR-10: The system shall store reservation details.

• FR-11: The system shall allow a Driver to cancel an eligible reservation.

• FR-12: The system shall allow a Driver to view booking history.

• FR-13: The system shall allow an Admin to manage reservations.

5.4 Check-in, Check-out and Charges

• FR-14: The system shall allow a Driver to check in for a parking session.

• FR-15: The system shall allow a Driver to check out from a parking session.

• FR-16: The system shall calculate the applicable parking fee during the check-out/payment workflow.

• FR-17: The system shall maintain the relevant parking session information.

5.5 Payment and Receipt

• FR-18: The system shall allow a Driver to make a parking payment.

• FR-19: The system shall send/process payment information through the Payment Gateway.

• FR-20: The system shall generate a digital receipt after payment processing.

• FR-21: The system shall maintain payment-related transaction information.

5.6 Notifications

• FR-22: The system shall provide notifications to the Driver for relevant parking activities.

5.7 Administration

• FR-23: The system shall allow an Admin to manage parking areas.

• FR-24: The system shall allow an Admin to manage parking slots.

• FR-25: The system shall allow an Admin to manage parking pricing.

• FR-26: The system shall allow an Admin to monitor parking occupancy.

• FR-27: The system shall provide parking statistics to the Admin.

• FR-28: The system shall provide revenue reports to the Admin.

5.8 IoT / ANPR Integration

• FR-29: The system shall receive vehicle entry/exit events from the Sensor/Camera (IoT/ANPR) component.

• FR-30: The system shall update parking occupancy based on detected vehicle entry/exit events.

6. Non-Functional Requirements

Attribute	Requirement
Performance	Common user operations such as parking search and availability viewing should respond within an acce
Security	User authentication and authorization shall protect account and administrative functionality. Sensitive d
Reliability	Reservation and occupancy data should remain consistent and should not result in conflicting slot state
Usability	The interface should provide clear navigation, understandable slot availability, meaningful validation, a
Maintainability	The system should be modular so that parking, reservation, payment, notification, administration, and I
Scalability	The architecture should allow additional parking areas, slots, and users to be added without major rede

7. Scope and Assumptions

The requirements in this report are derived from the supplied UML diagram. The diagram explicitly includes IoT/ANPR-based vehicle detection and an external Payment Gateway; these are therefore treated as part of the represented system scope.

The exact technology stack, database, payment provider, notification provider, ANPR implementation, and deployment environment are not specified by the UML diagram and should be finalized separately during system design.

No additional features have been added to the requirement list merely because they are common in parking applications; the report is intended to remain aligned with the supplied use case model.

8. Use Case to Requirement Traceability

Use Case	Related Requirements
Register / Login	FR-01, FR-02
Manage Profile	FR-03
Search Parking	FR-04
View Slot Details	FR-05
Check Parking Availability	FR-06, FR-07
Reserve Parking Slot	FR-09, FR-10
Cancel Reservation	FR-11
Check-in / Check-out	FR-14, FR-15, FR-17
Make Payment / Process Payment	FR-18, FR-19, FR-21
Generate Digital Receipt	FR-20
View Booking History	FR-12
Receive Notifications	FR-22
Admin Management	FR-13, FR-23–FR-28
Detect Vehicle Entry / Exit	FR-29
Update Parking Occupancy	FR-30

Appendix A — Provided UML Use Case Diagram

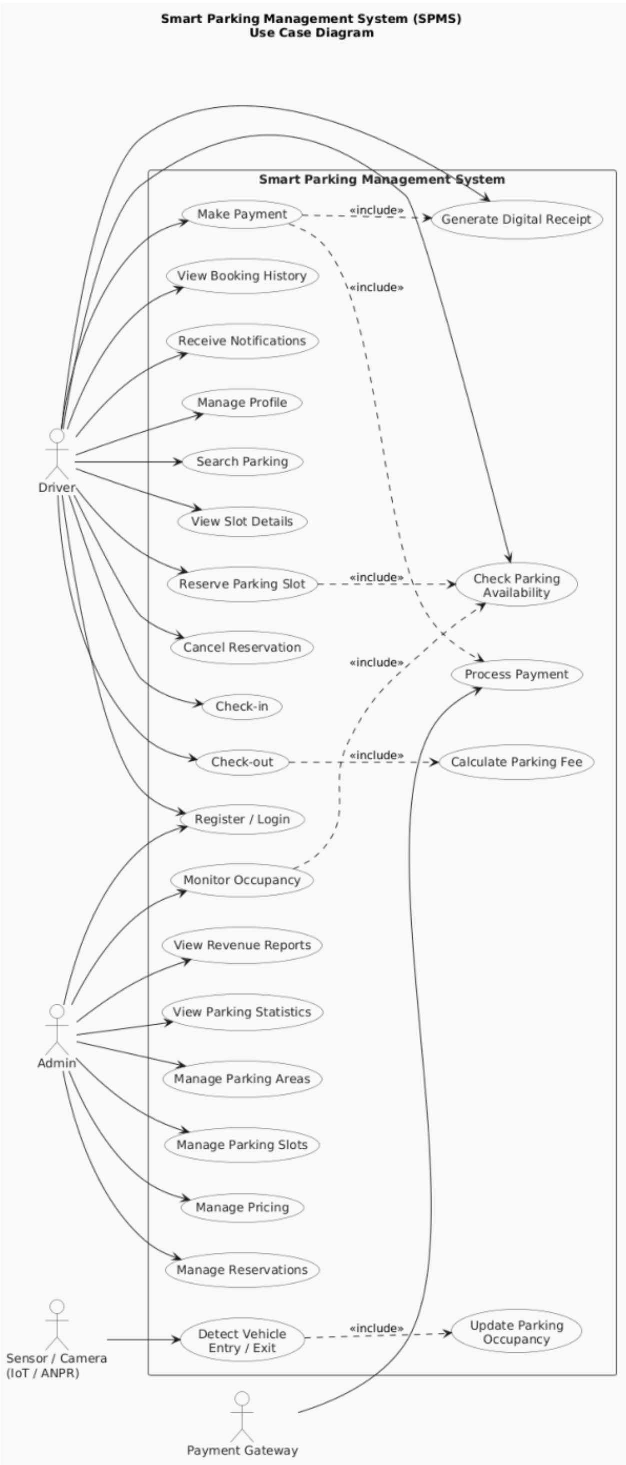

Figure A-1. Smart Parking Management System (SPMS) Use Case Diagram supplied for this report.